

Choosing a Deep-Research Method

A domain-agnostic protocol for an as-yet-unspecified system-design research task, selected from the current agentic-research literature under a hard zero-additional-spend constraint.

Scope research methodology only	Constraint \$0 marginal spend	Assumed nothing about domain, hardware, OS, workload	Compiled 30 Aug 2026
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01 Determination

Run a single-threaded iterative draft-refine loop, with depth-1 bounded fan-out for breadth-only retrieval and a rubric-driven verification pass in isolated context.

Backbone Draft-then-revise (TTD-DR pattern). One continuous context holds the evolving report; each round targets the draft's largest uncertainty, not a pre-planned section list.	Parallelism Permitted only for read-only, independent, breadth-first enumeration. One level deep. No subagent spawns a subagent.
Verification A separate pass in fresh context, scored against a written rubric, not a debate between peers. This is the highest-leverage component in the whole design.	Rejected as default Flat multi-agent swarms, multi-level hierarchy, competitive or consensus-seeking debate, autonomous spawn-on-demand adaptivity.

The determination is a hybrid because the evidence does not support any single family as a general winner. It supports a specific division of labour: sequential depth for reasoning, parallel breadth for enumeration, isolated critique for verification. The families below are graded against that division, not against each other in the abstract.

02 How the evidence was graded

The agentic-research literature is unusually contaminated by vendor self-report and by leaderboard aggregators that do not control for harness configuration. Every claim below carries a tier.

Tier A Peer-reviewed / conference ACL, ICLR, and equivalent venues. Controlled comparisons with stated budgets. Weighted highest, including where it contradicts vendor claims.	Tier B Preprint with released artifacts arXiv with public code, data, or annotator released. Most of the 2026 material sits here. Directionally trusted; single results not treated as settled.
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Tier C | Lab engineering publication

Anthropic, Google Research, LangChain, Cognition. High operational value, but internal evals are unreproducible. Used for failure modes and design constraints, not for magnitude claims.

Tier D | Third-party leaderboard aggregation

Used only to establish that a capability frontier exists. Two aggregators queried this session disagreed on whether a top BrowseComp score belonged to a single-agent or a four-way parallel configuration - which is precisely the variable under study.

Disqualifying caution

No public benchmark isolates architecture from model capability. Anthropic's own analysis found token expenditure alone explains roughly 80% of performance variance on browsing evaluations, and that upgrading the model beats doubling the token budget. Any headline "architecture X beats Y" number that does not hold model and tokens fixed is measuring spend, not design.

03 Architecture families, assessed

Single-agent	Matches or beats every multi-agent variant tested - sequential, subtask-parallel, parallel-roles, debate, ensemble - on multi-hop reasoning when thinking tokens are held constant, across Qwen3-30B, R1-Distill-70B and Gemini 2.5, at every budget above 100 tokens. Degrades only when context quality itself degrades.Tier B arXiv 2604.02460	1x	Backbone
Multi-agent (flat)	Anthropic reports +90.2% over single-agent Opus 4 on an internal research eval, for breadth-first queries requiring parallel directions - and in the same document reports ~15x chat token usage and explicit unsuitability where agents share context or interdepend. LangChain's Open Deep Research abandoned its supervisor/multi-agent pattern for a simpler agentic loop on performance grounds.Tier C Anthropic Engineering; langchain-ai/open_deep_research	~15x	Conditional
Hierarchical	No controlled study located that isolates depth as a variable. MAST's 1,600+ annotated traces across 7 frameworks give 14 failure modes in 3 categories - system design, inter-agent misalignment, task verification - and conclude that structural redesign alone is insufficient. Depth multiplies exactly the misalignment surface MAST measures.Tier B arXiv 2503.13657 (kappa=0.88)	15x+	Reject
Iterative (draft-refine)	The strongest single result in the corpus for long-form research output. Treating the report as a noisy draft denoised by retrieval yields a 74.5% win rate against OpenAI Deep Research on DeepConsult, +7.7% on HLE-Search, +1.7% on GAIA, with ablations attributing gains to the retrieval-denoising and self-evolution components separately.Tier B Google Research, TTD-DR (arXiv 2507.16075)	2-4x	Backbone
Recursive	Root agent orchestrates over a context variable rather than a context window. On OOLONG aggregation, 0.44 -> 0.79 at 128k tokens; the non-recursive agent refused the task outright at that length. Buys deterministic coverage - a loop guarantees every chunk is visited, model judgment does not.Tier B/C RLM; LangChain Deep Agents	variable	Conditional
Adversarial / debate	Protocol design decides the sign of the effect. Competitive debate degrades up to 15 points below plain chain-of-thought - debaters optimise persuasion, not truth. Consensus-seeking debate suppresses the informative disagreement it was meant to surface. Only explicitly collaborative, non-zero-sum protocols beat self-consistency at matched budget.Tier B arXiv 2510.20963; 2605.00914	3-8x	Reject as default

Verifier-based	The bottleneck, and the best return on spend. Parallel sampling raises pass@K ~50% from K=1 -> 4 while realised accuracy barely moves - a measured verification gap between having the right answer among samples and picking it. Rubric-guided test-time verification closes part of it: +12-48% verifier meta-F1, and 8-11% end-to-end accuracy on hard GAIA / Xbench subsets as a plug-and-play module with no retraining. Tier A ACL Findings 2026 (DeepVerifier); Tier B arXiv 2602.18998	1.3-2x	Adopt
Adaptive	Sound in principle, unvalidated in the specific form usually proposed. Sequential scaling hits a measurable context ceiling beyond which added history plateaus or degrades; parallel scaling stalls on the verification gap. So effort should adapt - but on measured signals. Autonomous spawn-on-demand is a documented Anthropic failure mode: early systems spawned excess subagents for simple queries and searched endlessly for sources that did not exist. Tier B/C arXiv 2602.18998; Anthropic Engineering	varies	Rule-based only
Emerging	Context engineering over agent count (share full traces, not messages); code-as-orchestration with results held in variables; open-weight research models - a 30B MoE web agent reports HLE 32.9 / BrowseComp 43.4 / xbench-DeepSearch 75, and ships both a native ReAct mode and a heavy test-time-scaling mode. Frontier BrowseComp scores now cluster within ~1.4 points, which makes harness design, not model choice, the remaining lever. Tier C Cognition; Tier B Tongyi DeepResearch; Tier D aggregators	-	Absorb principles

04 The protocol

Six phases. The order is load-bearing: each phase's gate is what makes the next phase's spend defensible. Phases 2 and 3 loop; everything else runs once.

P0

Fix the decision, not the answer

Before retrieval, write the question as a decision with named criteria and explicit unknowns. Because the domain is undetermined, the artifact of this phase is a parameterised criteria set - the axes on which any candidate design will be judged - plus the list of facts that would have to be true for each axis to matter. Nothing about purpose, hardware, workload, or OS is inferred to fill a blank; blanks are recorded as blanks.

Gate - Every criterion is falsifiable and every unknown is written down. An unstated assumption discovered later invalidates the phases built on it.

P1

Write the bad draft first

Produce a full-shape report from prior knowledge alone, with every unsupported claim tagged. This is the noisy initialisation the whole loop denoises. Its purpose is not to be right - it is to expose the shape of the ignorance so that retrieval is aimed rather than opportunistic. This inverts the common plan-then-search order and is the mechanism TTD-DR ablates as the source of its gains.

Gate - Draft covers every P0 criterion. Tagged claims outnumber supported ones; that is expected here.

P2

Retrieve against the largest gap

Each round: rank the draft's tagged claims by decision impact x uncertainty, retrieve against the top one only, revise the draft in place, re-rank. Single context, sequential. This is where the reasoning happens and where the single-agent result applies most directly - under equal token budgets nothing tested beats it on multi-hop chains.

Gate - Each round must change the draft. A round that retrieves without altering a claim or a confidence is a signal to re-rank, not to retrieve harder.

P3

Fan out for enumeration only

Where a gap is genuinely breadth-first - enumerate the candidate approaches, collect the same five properties for each of N options, survey what exists - dispatch parallel read-only workers. Three hard limits: depth 1, no worker spawns a worker; workers receive the full framing, not a one-line task, since incomplete context is Cognition's documented cause of conflicting implicit decisions; and every worker returns evidence to one synthesiser that owns all interpretation. Workers gather. They do not conclude.

Gate - Fan-out is justified in writing before dispatch, on the test "would sequential retrieval produce the same set more cheaply?" If yes, do not fan out.

P4

Verify in isolated context, against a rubric

The single highest-leverage phase. A fresh context receives the draft without the reasoning trace that produced it, and scores it against a written rubric: claim-to-source fidelity, source authority, criteria coverage, unstated assumptions, and internal contradiction. Isolation matters - the same context that produced a conclusion is a poor judge of it, and unguided peer debate is empirically worse than isolated self-correction. Findings return as defects to fix, not as votes to tally.

Gate - Zero unresolved critical defects. Each defect is closed by evidence, by a downgraded confidence, or by an explicit "unresolvable at zero cost" entry.

P5

Stop on measured saturation

Termination is a decision with criteria, not a feeling of doneness. Stop when: two consecutive P2 rounds yield no new entities or claim changes; every P0 criterion carries a supported position or a stated blank; and the P4 rubric passes clean. Two named failure modes are checked explicitly at this gate - under-retrieval (stopping before the obscure items surface) and hedging (padding the answer set with low-confidence items to inflate apparent recall), both measured as the dominant residual errors of state-of-the-art research agents.

Gate - Saturation is demonstrated, not asserted. If a criterion cannot be closed for free, it ships as an open item with the cost of closing it named.

05

Zero-cost execution

The constraint is not incidental - it eliminates the architecture the vendor literature most enthusiastically recommends. A ~15x token multiplier against a fixed subscription allowance is a real budget, just denominated in usage rather than dollars. The protocol above spends roughly 2-4x a baseline pass, concentrated in the phases with measured returns.

What the constraint permits

- Existing subscription research modes - already-paid deep-research features carry zero marginal cost. Running two different vendors' research modes over the same P0 brief is a heterogeneous ensemble for free; disagreement between them is a retrieval signal, and each becomes a legitimate P4 cross-check on the other.
- Free scholarly APIs - arXiv, OpenAlex, Semantic Scholar and equivalent open corpora cover the Tier A/B evidence base without a paid search key.
- Open-weight research agents - a 30B-class open web agent is downloadable at no licence cost, subject to local hardware. Treat as optional; it is a hardware question, and hardware is one of the things this brief refuses to assume.

What it rules out

- Persistent multi-agent swarms as the default execution mode.
- Metered search APIs beyond a permanent free tier - free trials do not satisfy the constraint.
- Large-K parallel sampling, which pays the full token cost to buy a pass@K improvement the selection step cannot cash in.

Controls against known failure modes

Conflicting implicit decisions	Parallel workers making incompatible assumptions with no view of each other	P3 depth-1 cap; full framing to every worker; single synthesiser owns all interpretation
Inter-agent misalignment	MAST category (ii) - one of 3 categories over 14 modes	Workers never negotiate; they return evidence to one owner
Task verification failure	MAST category (iii); the measured verification gap	P4 is a first-class phase with its own gate, not a final read-through
Context ceiling / degradation	Accumulated history past a threshold plateaus or degrades performance	Draft is the state, not the transcript; P2 revises in place rather than appending
SEO-authority inversion	Observed in production: agents prefer optimised content over authoritative sources	Source authority is an explicit P4 rubric axis; Tier A/B corpora preferred at retrieval time
Endless search for non-existent sources	Observed in production multi-agent research systems	P2 no-change rule and P5 saturation test both terminate on measured non-yield
Debate hacking	Overconfident rhetoric under competitive protocols; redundant echoing under consensus-seeking ones	No peer debate. Verification is rubric-scored in isolation.

What would overturn this

Stated so the determination is falsifiable rather than merely argued.

- The eventual task turns out to be enumeration-dominant - e.g. "survey every option in a large candidate space" with little cross-claim reasoning. Then P3 becomes the main phase and the flat multi-agent evidence, not the single-agent evidence, governs. This is the most likely overturn, and P0 is where it would be detected.
- A controlled study isolating architecture from model and token budget lands with a different result. None currently exists; the entire comparison above rests on partial controls.
- The verification gap closes in a released model. Large-K parallel sampling then becomes rational, and the cost calculus in Section 05 changes.
- Inputs exceed a single context by a wide margin - many long documents rather than many short retrievals. The recursive/code-orchestration pattern moves from conditional to primary.

Honest limitation

Most of the 2026 material cited is preprint, not peer-reviewed. Two of the load-bearing quantitative results - the +90.2% multi-agent figure and the 74.5% win rate - come from parties with an interest in the outcome, on evaluations they designed. They are used here for direction and for failure-mode identification, never as the sole basis for a phase. The verifier evidence, which carries the most weight in this determination, is the only Tier A item in the set.

Sources

- **Tier A** | **ACL Findings 2026** [Inference-Time Scaling of Verification: Self-Evolving Deep Research Agents via Test-Time Rubric-Guided Verification \(DeepVerifier\)](#) | arXiv
- **Tier B** | **arXiv 2503.13657** [Why Do Multi-Agent LLM Systems Fail?](#) - MAST taxonomy, 14 modes / 3 categories, 1,600+ traces
- **Tier B** | **arXiv 2604.02460** [Single-Agent LLMs Outperform Multi-Agent Systems on Multi-Hop Reasoning Under Equal Thinking Token Budgets](#)

- **Tier B** | [arXiv 2602.18998](#) Benchmark Test-Time Scaling of General LLM Agents - context ceiling, verification gap
- **Tier B** | [arXiv 2510.20963](#) When and Why Does Multi-Agent Debate Fail and Does It Really Underperform?
- **Tier B** | [arXiv 2605.00914](#) The Cost of Consensus: Isolated Self-Correction Prevails Over Unguided Homogeneous Multi-Agent Debate
- **Tier B** | [arXiv 2605.01566](#) Multi-Agent Reasoning Improves Compute Efficiency: Pareto-Optimal Test-Time Scaling - the counter-case
- **Tier B** | [arXiv 2604.16529](#) Scaling Test-Time Compute for Agentic Coding - representation, selection, reuse
- **Tier B** | [arXiv 2601.20975](#) DeepSearchQA: Bridging the Comprehensiveness Gap for Deep Research Agents - under-retrieval, hedging
- **Tier B** | [arXiv 2506.11763](#) DeepResearch Bench - RACE / FACT evaluation frameworks
- **Tier B** | [arXiv 2506.18096](#) Deep Research Agents: A Systematic Examination and Roadmap - static vs dynamic workflow taxonomy
- **Tier C** | [Google Research](#) Deep Researcher with Test-Time Diffusion (TTD-DR)
- **Tier C** | [Anthropic Engineering](#) How we built our multi-agent research system
- **Tier C** | [Cognition](#) Don't Build Multi-Agents - context engineering principles
- **Tier C** | [LangChain](#) Open Deep Research - documented retreat from the supervisor pattern | RLMs in Deep Agents
- **Tier B** | [Tongyi Lab](#) Tongyi DeepResearch - open-weight 30B MoE web agent, ReAct and heavy modes
- **Tier D** | [aggregators](#) Steel BrowseComp leaderboard | BenchLM BrowseComp - mutually inconsistent on harness attribution

Scope boundary

Methodology selection only - no system design, no domain assumptions. Determination is conditional on P0 confirming a reasoning-dominant rather than enumeration-dominant task.